Kinds of collisions

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Learning target: Separate momentum conservation from kinetic-energy conservation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 3.*
$J=30$ N·s right; $v_f=10$ m/s right.

02. Worked example: explain the model
Two separate carts before; joined pair after at 2 m/s in the original direction.

03. Read and interpret the model
Joined carts share final velocity.

04. Apply the physics
2 m/s in original moving cart’s direction.

05. Explain your reasoning
Kinetic energy decreases but total energy is conserved; deformation, heat and sound may gain energy.

06. Review: connect an earlier idea *Review: Unit 2, Day 10.*
35 m from the base in the launch direction.

07. Exit check
Sticking; total momentum.

08. STRETCH: test the model
Yes, both totals 4 kg·m/s. Additional information about the interaction/energy is needed.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 3.*
J=7.5 N·s; Δv=12.5 m/s in the force direction.

02. Worked example: explain the model
Two separate carts before; joined pair after at 2 m/s in the original direction.

03. Read and interpret the model
Joined carts share final velocity.

04. Apply the physics
Total initial momentum = 6 kg·m/s, combined mass = 3 kg, v_f = +2 m/s. Initial K = 9 J; final K = 6 J. Transformed energy = 3 J, or one third of the initial kinetic energy.

05. Explain your reasoning
Kinetic energy decreases but total energy is conserved; deformation, heat and sound may gain energy.

06. Review: connect an earlier idea *Review: Unit 2, Day 10.*
t=3 s; range 18 m; vertical velocity 30 m/s downward.

07. Exit check
Initially moving mass stops; other moves atu, for ideal 1 D equal-mass elastic collision.

08. STRETCH: test the model
All give total 4 kg·m/s for equal masses. Only (2, 2) is a shared sticking velocity; momentum alone does not classify all interactions.